# Supplementary Material

## Where the Bits Must Live: A Tight Input-Side Information Bound for Fixed-State Spectral Models

Anonymous submission. This document contains the appendices to the main paper. It is self-contained: each proof restates the final premises and the identifier of the claim it establishes, and proofs cross-reference those identifiers rather than referring to 'the proof above'. Numerical witnesses certify non-vacuity and agreement; they are never used as premises.

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The main paper states four core results. SE1-SE5 in Appendix A.7-A.11 are supporting statements about an exactly-defined finite consolidation process; P1-P8 and L1-L3 are propositions and lemmas. No theorem is introduced in the supplementary material.

# Appendix A Complete Analytic Proofs

All proofs are analytic. Numerical witnesses certify non-vacuity and agreement with the statements; they are not used as premises.

## A.0 Notation and unit ledger

| symbol | meaning | unit / where used |
|---|---|---|
| B | total input-side bit budget, B = sum_t b_t | bits; Theorem A |
| b_t | per-position alphabet excess log2\|pi^(-1)(v_t)\| | bits; Section 2 |
| b | bits per insight per consolidation round | bits; CT7(b), SE4; here b = 5 = log2 V_K |
| w, T | insights per round, number of rounds | count; CT7(b), SE4 |
| state bytes | structural size of the recurrent state | BYTES, not bits; Table 8, Appendix D |
| 32768 | d x d fp64 core state, d = 64 (fusion config) | bytes; Tables 16, 18, 34; DIFFERENT from Table 8 |
| c, o | orthogonal successor rank per hop, tensor order | count; P5, Result 2 |

**Table A0.** Uppercase B and lowercase b are different quantities; state bytes is a size in bytes and never a bit budget. Table 8 (throughput configuration) and the fusion tables (d = 64) use DIFFERENT state configurations and must not be compared numerically.

## A.1 Theorem A(a) and A(b)

Fix a fibre v. The decoder observes (v, s) with s in (0,1)^B, so F(v, .) has at most 2^B distinct values. Let A(v) be the set of distinct answers over instances compatible with v and in the support. If |A(v)| > 2^B, two instances G, G' in the fibre share (v, s) but have k_H(G) != k_H(G'). F cannot equal both. Under the uniform measure on the fibre, at most 2^B of the |A(v)| answer classes can be correct, hence acc(F | v) <= 2^B / |A(v)|; this proves (a). The <= 1/2 corollary follows only when 2^B <= |A(v)|/2, i.e. B <= log2|A(v)| - 1. Exactness on every instance requires |A(v)| <= 2^B for all v, i.e. B >= log2 |A(v)|; this proves (b). Neither step uses A3. The marginal accuracy E_v[acc(F|v)] is NOT bounded by 1/|A| when fibres with |A(v)|=1 carry most of the mass, which is why Section 4 labels every statistic. QED.

Binary-chain witness for |A| = 2^m. Fix m <= n and H = m. At hop h the visible flat sequence contains one key k_h and TWO candidate successor symbols; a bit d_h not recorded in v selects which one is bound, and the selected symbol is the key of hop h+1. Both candidates are present as tokens, so every d in (0,1)^m yields the SAME v and the same q = k_1. Map d to the endpoint id sum_h d_h 2^(h-1); with V_K >= 2^m these are 2^m distinct endpoints. The witness enumerates all 2^m assignments with bit-identical visible inputs (max deviation 0.0e+00). If the pairing is also absent from v, all n! matchings belong to the fibre, giving B >= log2(n!). QED.

## A.2 Theorem A(c) and the Toeplitz equivalence (P6)

In class M, layer h computes z_h = sum_t K_h(L-1-t) m_h[t] e(x_t) with m_h[t] = <q_h, e(x_(t-1))>, and q_(h+1) = T_h z_h / ||T_h z_h||. K_h depends only on the lag and on fixed parameters, T_h is fixed, and e is fixed. The composition is therefore a function of the multiset of (flat symbol id, position) pairs plus q, i.e. of v with the refinement s removed. Under arbitrary filler spacing the positions of the pair tokens are determined by filler counts independent of the role assignment, so F is constant on the fibre: no choice of H_L, state order or (rho, theta, w) changes this, and A(a) applies. To encode roles in the kernel one makes the recurrence coefficients functions of x_t, hence time-varying. For y_t = a_t y_(t-1) + b_t y_(t-2) + c_t u_t with c_t != 0, P6 gives h(s,s)=c_s, h(s+1,s)=a_(s+1) c_s, h(s+2,s)=(a_(s+2)a_(s+1)+b_(s+2)) c_s. Toeplitz forces c_s, a_s and b_s constant by induction on the lag, so (rho_t, theta_t) are constant. Non-Toeplitz h(t,s) has no single H(z) and the single-FFT path fails. This negates THAT path within M, not every conceivable O(L log L) algorithm. A(c) is therefore a conditional statement: IF duality is preserved, the bits must enter on the input side. QED.

## A.3 Theorem A(d), A(e), A(f) and P2

A(d). By the Section 2 definition B = sum_t log2 |pi^(-1)(v_t)|, so an alphabet partition with |pi^(-1)| = 2 contributes exactly one bit per non-empty token and is COUNTED in B. An explicit tag and a symbol id carrying a role are the same quantity in a different carrier. Tagging only the KEY token of each of n pairs gives B = n, which attains B >= n from A(b). A positional prior is a support restriction: it reduces |A(v)| rather than increasing B, and under random filler spacing the restriction no longer holds, so the effective information falls to zero (measured 0.3438, 5 seeds). QED.

P2 (explicit construction). Fix keys and q; A = sum_p L_p v_p + c with d x d blocks L_p, read-out f(A,q) arbitrary, target j = j(q). Case 1: L_j singular. Pick nonzero delta in ker L_j and change only v_j; delta-A = 0, answer changes. Case 2: L_j invertible and some L_p != 0 for p != j. Pick delta_p and set delta_j = -L_j^(-1) L_p delta_p; then delta-A = L_p delta_p + L_j delta_j = 0, and delta_j is nonzero, so the target value changes. Case 3: L_j invertible and all other L_p = 0. Then A depends only on v_j; fix all values and redirect q to a different pair p != j, which changes the required answer while A is unchanged. In every case two legal assignments give the same (A,q) and different answers, so f cannot be exact. No step infers a target-coordinate perturbation from dim ker L > 0 alone. Scope: single mixer with a fixed linear historical summary; deep stacks with per-position nonlinear FFNs, input-dependent gates and outer-product states leave the premise. QED.

A(e). B < log2|A| makes exactness impossible for EVERY deterministic model by A(b), hence for every H_L; for B >= log2|A| Result 2(ii), under its stated premises, gives minimum depth H. The frontier is a right angle. A(f): under A4 the decoded value at depth H_L is k_(H_L); for H_L > H and distinct keys this differs from k_H. Retaining and selecting the H-th intermediate output gives the correct answer for any H_L >= H; the lower bound H_L >= H is unchanged. QED.

## A.4 Result 2: induction and tightness

(i) P2 rules out an order-1 vector state with a linear summary; Construction C gives an order-2 tensor state with a contraction read-out, so the minimum is exactly 2. (ii) Induction on H. Base H=1: one contraction resolves one content match and yields the value; a zero-th layer cannot. Step: suppose H-1 layers resolve H-1 hops and produce q_H = C[k_H] by the unique H6 interface. Layer H performs one further match with q_H and returns k_H's successor. Fewer than H layers leave at least one hop unresolved: the last read-out sees a mixture of at least two successors and cannot be exact on all fibres. The analytic construction attains 1.0000 on the diagonal of the 5 x 5 grid and <= 0.10 off it, so the bound is tight and non-vacuous. Overshoot H_L > H is A(f), not a failure of the lower bound. QED.

## A.5 Proposition P5: conditional state-order lower bound

Premise. The interface is H6 with rank at most D; the codebook rows are mutually orthogonal; at each of H hops there are c mutually orthogonal candidate successors; answers are determined by the joint choice of the start key and the H successors, so the candidate set has size c^(H+1). Claim. An order-o tensor state whose per-layer linear summary has image dimension at most o*d cannot be exact unless o >= H+1. Proof. The H+1 independent matching directions must be simultaneously recoverable from the summary for every one of the c^(H+1) joint choices. If o <= H, the summary is a linear map from a space of dimension (H+1)d into one of dimension o*D <= H*D; its kernel has dimension at least d > 0. By the P2 construction (with the blocks taken to be the coefficient matrices of the H+1 directions) there exist two distinct joint choices producing the same summary and the same q, hence the same read-out, yet requiring different answers. Exactness is impossible; therefore o >= H+1. This is a CONDITIONAL bound: when each hop offers only 2 candidates the same counting gives 2^(H+1) classes and the necessary order is weaker, degenerating to a log2(H+1) bit requirement. We state that degeneration explicitly rather than presenting c^(H+1) as unconditional. QED.

## A.6 P1, P3, P4, P7, P8 and non-vacuity

P1. H(z) = sin(theta) / (1 - 2 rho cos(theta) z^(-1) + rho^2 z^(-2)) gives y_t = 2 rho cos(theta) y_(t-1) - rho^2 y_(t-2) + sin(theta) u_t, a real-field equivalence; different rounding orders in the FFT and IIR paths make fp32 bitwise identity false (W1). P3. For a finite output set F with |F| <= |S|, the state can encode f(x) directly, so the literal claim 'O(1) states must lose' is false; the correct statement is that a SPECTRAL compression state loses content addressing. P4. At rho=1 and a single pole the Construction C state reduces to sum_tau k_tau v_tau^T, the linear-attention state; rho<1 retains

a decay spectrum. P7. With impulse response bounded by rho^lag / |sin(theta)| and per-step injection bounded by delta_max, the output error is at most delta_max / ((1-rho)|sin(theta)|), under a uniform floating-point perturbation model. P8. As theta_m -> 0 the dictionary columns approach theta_m rho^lag (lag+1) and become collinear, so the grid is asymptotically rank deficient. Non-vacuity: n=4, m=2, V_K=32, H=2, flat alphabet, arbitrary filler spacing give |A|=4 at B=0 with acc <= 1/4, and exactness at B=2.

### A.7 SE1: orthogonal-registration zero-crosstalk identity

PREMISE. C in R^(V x V) satisfies C C^T = I; S = sum_(a=1)^m w_a C[k_a]^T C[v_a] with distinct k_a; the query is q = C[k]. CLAIM. C[k] S = w_j C[V_K + v_j] when k = k_j, and 0 when k is not among the k_a. PROOF. C[k] C[k_a]^T = delta_(k,k_a) by row orthogonality, so C[k] S = sum_a w_a delta_(k,k_a) C[V_K+v_a]. Distinctness leaves at most one nonzero term. CONSEQUENCE. The read-out logits = (C[k] S) C[V_K:,:]^T are a one-hot vector scaled by w_j for a registered key, so accuracy is IDENTICALLY the fraction of correctly registered bindings. There is no cross-talk term to bound or estimate. NON-VACUITY. V = 64, V_K = 32, C the normalized Sylvester matrix, m >= 2: 200 randomized instances give a maximum on-target error of 4.4e-16 and maximum off-target error of 8.5e-17. BOUNDARY. Repeated keys, a rank-deficient codebook, or a non-codebook registration vector break the identity; this is why the break-alignment arm collapses to chance.

### A.8 SE2: monotone improvement and cover time

PREMISE. n bindings arrive by i.i.d. sampling with replacement (coupon collector); the gate has false-accept at most alpha and miss at most beta; accepted experiences are correct; rho < 1 with a precision floor epsilon such that SE5's capacity K* >= n. CLAIM. (i) the number m_t of correct registrations is monotone non-decreasing; (ii) E[m_T] >= (1-beta) T until saturation; (iii) E[T_cover] = n H_n / (1-beta) with H_n = sum_(i=1)^n 1/i; (iv) Pr(T_cover > (1+eps) E T) <= exp(-eps^2 (1-beta) n H_n / 3); (v) the probability of any false commit in T rounds is at most min(1, T alpha). PROOF. (i) SE1 makes each correct registration a disjoint orthogonal term; register() replaces only the term of the same key and never removes another key's correct term, and K* >= n prevents decay below epsilon. (ii) The gate accepts a true binding with probability at least 1-beta. (iii) The classical coupon-collector expectation n H_n divided by the effective arrival probability 1-beta. (iv) Chernoff on a sum of n geometric variables, equivalently the union bound Pr(some type missing) <= n(1-1/n)^T. (v) Boole's inequality over T rounds. NON-VACUITY AND CHECK. For n = 32, alpha = 0.05, beta approximately 0, the prediction is 129.87 rounds; the measured five-seed mean first full-cover round is 130.00 (relative gap 0.10 percent), with per-seed values 170, 100, 90, 120, 170. With n = 5 the sample SD is 38.08 and the 95% t-CI is [82.72, 177.28] rounds: the prediction lies inside the interval, but the interval spans 73% of the mean, so this is weak evidence and the small gap is NOT a precision claim. The arrival model was set to i.i.d. BEFORE measuring. BOUNDARY. A deterministic round-robin schedule gives T_cover = n and SE2 does not apply; large beta stretches the cover time by 1/(1-beta); if K* < n then SE5 takes over and full cover is impossible.

### A.9 SE3: the statistical gate is necessary

PREMISE. Each accepted experience is corrupted independently with probability p; a corrupted value is uniform on [0, V_K); there is no gate; the final value of a key is its last write. CLAIM. E[acc] = (1-p) + p/V_K = 1 - p(1 - 1/V_K) exactly, and E[acc] <= that value in general because some keys may never arrive. PROOF. The last write is correct either when it is not corrupted (probability 1-p) or when corruption happens to draw the true value (probability p/V_K). SE1 makes the keys independent, so accuracy is the mean of n such Bernoulli variables. CONCENTRATION. Over N independent key-level trials, Hoeffding gives Pr(|a_bar - mu| >= t) <= 2 exp(-2 N t^2). With N = 640000 the 4-sigma interval is [0.7071, 0.7116] and the enumeration mean is 0.7091, inside it. WHY NECESSARY. Orthogonality prevents cross-talk but cannot prevent a wrong value from occupying an orthogonal slot; hence without a gate SE2's full cover is unattainable under corruption. SCOPE. With p = 0, mu = 1 and the gate is irrelevant: the ungated clean arm ties the fused arm exactly. The gate is necessary ONLY when experiences can be wrong. W15. An earlier draft called mu an almost-sure upper bound and tested it pointwise against a sample mean; a sample mean legitimately exceeds an expectation. The claim is now an exact expectation with a concentration interval.

### A.10 SE4: compatibility with the CT7 rate bound

CT7(b) bounds the information that T consolidation rounds can write into the state by T w b bits, independent of context length L and of bank capacity C. SE2 permits about (1-beta) T registrations and a corresponding accuracy gain. CLAIM. Both hold together. PROOF. Registration is injective (SE1's partial isometry), so one binding corresponds to one key-value association of exactly log2(V_K) bits. With T = 100 and w = 1, SE2 registers 100 bindings using 500 bits, while CT7 allows 500 bits: equality, not violation. MEANING. Self-evolution is real but bounded. Long context buys inference-time retrieval, not a faster learning rate. CT7(c)'s earlier statement that improvement was not proved is resolved for this finite process; CT7(b)'s upper bound is untouched.

### A.11 SE5: optimal decay eviction and closed-form capacity

PREMISE. Slot weights w_a = rho^age(a) with rho < 1; utility is additive and orthogonal (SE1); precision floor epsilon; capacity C_max. CLAIM. (i) Greedy removal of the smallest weight is optimal; (ii) the number of slots whose weight stays above epsilon is K* = floor(log epsilon / log rho). PROOF. (i) Utility U = sum_a w_a u_a with u_a >= 0 and mutually independent; retaining the C_max largest weights maximizes U, and dropping the smallest is its online realization. (ii) rho^a > epsilon iff a < log epsilon / log rho. NON-VACUITY. At rho = 0.999 and epsilon = 1e-3, K* = 6904; at rho = 0.90, K* = 65. The fast-decay arm loses accuracy to 0.1313 while the standard arm stays at 1.0000 over 320 rounds, matching the predicted capacity collapse. BOUNDARY. rho = 1 is critically stable, non-BIBO and has amplitude growing with L; the architecture excludes it and no run instantiates it. A near-critical value such as 0.99999 barely decays within the run and therefore has no discriminating power; 0.90 was chosen so SE5 is observable.

## Appendix B Withdrawn Statements

| ID | Withdrawn statement and reason |
|---|---|
| W1 | Unconditional fp32 bitwise FFT/IIR duality: different rounding orders; real-field equivalence only. |
| W8 | Content addressing requires input-dependent kernel coefficients: refuted by Construction C with strictly static kernels. |
| W9 | Absolute-position parity is a general role basis: a hidden format prior; collapses under arbitrary spacing. |
| W10/W11 | 'Exactly one bit per token is necessary and sufficient': the necessary quantity is total B >= log2\|A\|; one bit per token is a sufficient carrier. |
| W12 | Single-hop attention parity (Fisher p=0.4998): the stored baseline did not reproduce in the same-run audit (Section 4.6). No equivalence or superiority claim is made. |
| W13 | Trained corroboration of the layer bound under the earlier protocol: withdrawn after the fairness audit removed two input asymmetries. Re-established on the corrected architecture in Section 4.12; the earlier number is not reused. |
| W14 | Straight-through inter-layer normalisation (norm taken outside the graph): finite-difference gradient deviation 2.5e-03. Replaced by a differentiable square root built from log and exp, deviation 2.9e-10; all conservation checks are rerun on final weights. |
| W15 | SE3 as an almost-sure upper bound tested pointwise against a sample mean: a finite sample mean legitimately exceeds an expectation. Replaced by the exact expectation 1-p(1-1/V_K) with a Hoeffding concentration interval (Appendix A.9). |

## Appendix C Reproducibility

The submission is accompanied by an anonymised artifact containing the reference implementation of the layer, the seven same-run mechanism references used only for the Section 4.6 audit, the reachability enumeration of Section 4.3, and all raw result files. Every number in Tables 2-4 is injected from those files at build time; no value is transcribed by hand. All probes are CPU-only NumPy and complete in minutes. The Section 4.12 run uses 1500 Adam steps, lr 0.02, batch 64, five seeds, and 512 training plus 512 held-out instances per seed; the architecture module and every

convergence curve are in the artifact. The fusion run of Section 4.13 and Appendices D-E uses 320 i.i.d. arrival rounds, 32 bindings, five seeds, rho = 0.999 and epsilon = 1e-3; all seven arms, sixteen ablation cells and every individual seed are printed in Appendix D. Seeds are listed per table. The environment, dependency versions and a one-command reproduction script are in the artifact. The 2027 call for papers and submission deadline were not published at the time of writing; page count, the author-side LLM-use disclosure and reciprocal-reviewer obligations must be rechecked against the official template when it appears.

No count of statements is presented as a contribution. The withdrawn-statement register in Appendix B is a soundness practice, not an originality claim.

## Appendix D  All Raw Fusion Measurements

Every row below is a single seed of a single arm. Nothing is averaged away and no value is transcribed by hand: the generator reads the raw result files at build time. Protocol: 32 bindings, 320 rounds of i.i.d. arrival, rho = 0.999 unless stated, precision floor epsilon = 1e-3, capacity 64, one-sided paired-CI gate with alpha = 0.05, tau = 0, 8 probes and noise sigma = 0.10. Chance is 1/32 = 0.0313. The core state is 32768 bytes in every row (d = 64, fp64; notation table in Section 2). A machine-readable long-form CSV of every row below is supplied in the artifact as appendix_D_fusion_long.csv; no value is OCR-transcribed.

| arm | seed | final acc | commits | first full cover |
|---|---|---|---|---|
| fused gated | 0 | 1.0000 | 49 | 170 |
| fused gated | 1 | 1.0000 | 55 | 100 |
| fused gated | 2 | 1.0000 | 47 | 90 |
| fused gated | 3 | 1.0000 | 53 | 120 |
| fused gated | 4 | 1.0000 | 54 | 170 |
| no consolidation | 0 | 0.0000 | 0 | none |
| no consolidation | 1 | 0.0000 | 0 | none |
| no consolidation | 2 | 0.0000 | 0 | none |
| no consolidation | 3 | 0.0000 | 0 | none |
| no consolidation | 4 | 0.0000 | 0 | none |
| ungated clean | 0 | 1.0000 | 320 | 170 |
| ungated clean | 1 | 1.0000 | 320 | 100 |
| ungated clean | 2 | 1.0000 | 320 | 90 |
| ungated clean | 3 | 1.0000 | 320 | 120 |
| ungated clean | 4 | 1.0000 | 320 | 170 |
| ungated corrupt | 0 | 0.6250 | 320 | none |
| ungated corrupt | 1 | 0.7500 | 320 | none |
| ungated corrupt | 2 | 0.5625 | 320 | none |
| ungated corrupt | 3 | 0.5625 | 320 | none |
| ungated corrupt | 4 | 0.7188 | 320 | none |
| random write | 0 | 0.6250 | 154 | none |
| random write | 1 | 0.6875 | 153 | none |
| random write | 2 | 0.5938 | 154 | none |
| random write | 3 | 0.7500 | 155 | none |
| random write | 4 | 0.7812 | 155 | none |
| break alignment | 0 | 0.0000 | 49 | none |
| break alignment | 1 | 0.0000 | 55 | none |
| break alignment | 2 | 0.0938 | 47 | none |
| break alignment | 3 | 0.0000 | 53 | none |
| break alignment | 4 | 0.0312 | 54 | none |
| rho fast | 0 | 0.1562 | 49 | none |
| rho fast | 1 | 0.1562 | 55 | none |
| rho fast | 2 | 0.0938 | 47 | none |
| rho fast | 3 | 0.1250 | 53 | none |
| rho fast | 4 | 0.1250 | 54 | none |

**Table 18.** All 35 arm-seed runs. The first-full-cover column reports the first sampled round at which accuracy reaches 1.0 (sampling every 10 rounds); '-' means no such round occurred. The five fused values are the ones averaged in Section 4.13.

| corruption | config | mean | sd | n |
|---|---|---|---|---|
| 0.00 | gate off | 1.0000 | 0.0000 | 5 |
| 0.00 | alpha 0.10 | 1.0000 | 0.0000 | 5 |
| 0.00 | alpha 0.05 | 1.0000 | 0.0000 | 5 |
| 0.00 | alpha 0.01 | 1.0000 | 0.0000 | 5 |
| 0.00 | tau positive | 1.0000 | 0.0000 | 5 |
| 0.00 | sigma 0.30 | 1.0000 | 0.0000 | 5 |
| 0.00 | rho 0.90 | 0.1313 | 0.0234 | 5 |
| 0.00 | rho 0.999 | 1.0000 | 0.0000 | 5 |
| 0.30 | gate off | 0.6438 | 0.0781 | 5 |
| 0.30 | alpha 0.10 | 1.0000 | 0.0000 | 5 |
| 0.30 | alpha 0.05 | 1.0000 | 0.0000 | 5 |
| 0.30 | alpha 0.01 | 1.0000 | 0.0000 | 5 |
| 0.30 | tau positive | 1.0000 | 0.0000 | 5 |
| 0.30 | sigma 0.30 | 1.0000 | 0.0000 | 5 |
| 0.30 | rho 0.90 | 0.1000 | 0.0364 | 5 |
| 0.30 | rho 0.999 | 1.0000 | 0.0000 | 5 |

**Table 19.** All 16 corruption x configuration cells, each over five seeds. Under zero corruption the six gate configurations saturate and only the fast-decay configuration degrades; under 30 percent corruption the gate-off row degrades while every gated row saturates. This is the measured form of SE3's scope.

| check | pass |
|---|---|
| theory 7of7 | yes |
| theory SE1 | yes |
| theory SE2 | yes |
| theory SE3 | yes |
| theory SE4 | yes |
| theory SE5 | yes |
| theory nonvacuous SE2 | yes |
| theory SE3 concentration | yes |
| theory SE3 exact expectation | yes |
| evolution all invariants | yes |
| evolution monotone | yes |
| fused above 0.90 | yes |
| all key paired positive | yes |
| core state bytes constant across rounds | yes |
| slot count bounded by capacity | yes |
| SE5 capacity predicts rho fast loss | yes |
| ablation clean gate configs saturate | yes |
| ablation noisy gate necessary | yes |
| ablation rho decay matches SE5 | yes |
| SE2 prediction matches measurement | yes |
| SE3 concentration holds | yes |
| SE3 enumeration within 4sigma | yes |
| no gradient through retrieval | yes |
| no NxN score | yes |
| no kv cache | yes |
| v64 stack invariants | yes |
| v64 interface identities | yes |

**Table 20.** All 27 consistency checks on the FINAL post-evolution state. They cover the five analytic results, the paired statistics, monotonicity, core-state constancy, capacity bounds, the SE2 prediction-measurement agreement, SE3's concentration interval, the v6.4 stack invariants and the interface identities. A false here would be a finding, not a crashed run.

## Appendix E  Premise, Boundary and Non-Vacuity Ledger

A theorem whose premises admit no instance is empty. The table below gives one concrete satisfying instance, the exact boundary at which the statement stops, and what would falsify it, for every new supporting result. These are not additional contributions; they are the audit trail that keeps the premises honest.

| result | satisfying instance | boundary / falsifier |
|---|---|---|
| SE1 | V=64, V_K=32, Sylvester C, m>=2 distinct keys | repeated keys or rank-deficient code; any nonzero off-target read |
| SE2 | n=32 i.i.d. arrivals, beta~0, K*=6904 >= n | round-robin arrival; beta>0; K*<n; a non-monotone fused curve |
| SE3 | p=0.30, uniform corrupted value, no gate | p=0 makes the gate irrelevant; a gated arm below the exact expectation |
| SE4 | T=100, w=1, b=log2(32)=5, 500 bits used = 500 allowed | any run writing more than T w log2(V_K) bits into the state |
| SE5 | rho=0.999, eps=1e-3, K*=6904; rho=0.90, K*=65 | rho=1 is non-BIBO and excluded; a better-than-greedy eviction |
| CT7 | answer() is no-grad; selection Jacobian vanishes a.e. | a differentiable exact-retrieval channel through the online path |

**Table 21.** Each row names a measurement that would have to come out differently. The five supporting results SE1-SE5 do not change the four core results of Section 1.1; they make the empirical claim in Result 4 auditable at the level of individual runs. The withdrawn-statement register (Appendix B) records W13, W14 and W15, all of which are premise corrections rather than numerical adjustments.

## Appendix F  All 1500-Step Convergence Curves

Section 4.12 reports only the diagonal and the first-crossing step. Here every sampled point of every arm is printed for all five seeds, followed by every individual cell of the layer-by-hop matrix. Protocol: 1500 Adam steps, lr 0.02, batch 64, 512 training and 512 held-out instances per seed, width d = 64, fixed Hadamard embedding, evaluation every 150 steps. The architecture module's inter-layer map is the fixed codebook value-to-key swap; only the query projection and read-out are learned, from random

initialisation.

| arm | step | mean acc | sd | n seeds |
|---|---|---|---|---|
| arch codebook M | 0 | 0.0266 | 0.0064 | 5 |
| arch codebook M | 150 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 300 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 450 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 600 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 750 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 900 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 1050 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 1200 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 1350 | 1.0000 | 0.0000 | 5 |
| arch codebook M | 1500 | 1.0000 | 0.0000 | 5 |
| arch analytic witness | 0 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface random | 0 | 0.0301 | 0.0052 | 5 |
| legacy learnable interface random | 150 | 0.2812 | 0.0368 | 5 |
| legacy learnable interface random | 300 | 0.2633 | 0.0381 | 5 |
| legacy learnable interface random | 450 | 0.2492 | 0.0319 | 5 |
| legacy learnable interface random | 600 | 0.2398 | 0.0278 | 5 |
| legacy learnable interface random | 750 | 0.2383 | 0.0388 | 5 |
| legacy learnable interface random | 900 | 0.2422 | 0.0276 | 5 |
| legacy learnable interface random | 1050 | 0.2445 | 0.0321 | 5 |
| legacy learnable interface random | 1200 | 0.2520 | 0.0329 | 5 |
| legacy learnable interface random | 1350 | 0.2535 | 0.0311 | 5 |
| legacy learnable interface random | 1500 | 0.2492 | 0.0325 | 5 |
| legacy learnable interface fixedM | 0 | 0.0387 | 0.0119 | 5 |
| legacy learnable interface fixedM | 150 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 300 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 450 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 600 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 750 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 900 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 1050 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 1200 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 1350 | 1.0000 | 0.0000 | 5 |
| legacy learnable interface fixedM | 1500 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 0 | 0.0238 | 0.0106 | 5 |
| legacy trainable near-swap | 150 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 300 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 450 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 600 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 750 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 900 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 1050 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 1200 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 1350 | 1.0000 | 0.0000 | 5 |
| legacy trainable near-swap | 1500 | 1.0000 | 0.0000 | 5 |

**Table 22.** All 45 arm-step cells. The random learnable interface stays in 0.21-0.29 for the entire budget (this task's chance is 1/32 = 0.0313; 0.25 belongs to a DIFFERENT historical task); the codebook-fixed interface and the near-swap basin cross 0.99 by step 150 on every seed. The analytic witness is exact at step 0 by construction.

| hops | layers | seed | acc | params | steps |
|---|---|---|---|---|---|
| 1 | 1 | 0 | 1.0000 | 6144 | 1500 |
| 1 | 1 | 1 | 1.0000 | 6144 | 1500 |
| 1 | 1 | 2 | 1.0000 | 6144 | 1500 |
| 1 | 1 | 3 | 1.0000 | 6144 | 1500 |
| 1 | 1 | 4 | 1.0000 | 6144 | 1500 |
| 1 | 2 | 0 | 0.0215 | 6144 | 1500 |
| 1 | 2 | 1 | 0.0430 | 6144 | 1500 |
| 1 | 2 | 2 | 0.0332 | 6144 | 1500 |
| 1 | 2 | 3 | 0.0312 | 6144 | 1500 |
| 1 | 2 | 4 | 0.0273 | 6144 | 1500 |
| 1 | 3 | 0 | 0.0352 | 6144 | 1500 |
| 1 | 3 | 1 | 0.0410 | 6144 | 1500 |
| 1 | 3 | 2 | 0.0312 | 6144 | 1500 |
| 1 | 3 | 3 | 0.0312 | 6144 | 1500 |
| 1 | 3 | 4 | 0.0566 | 6144 | 1500 |
| 1 | 4 | 0 | 0.0469 | 6144 | 1500 |
| 1 | 4 | 1 | 0.0410 | 6144 | 1500 |
| 1 | 4 | 2 | 0.0352 | 6144 | 1500 |
| 1 | 4 | 3 | 0.0312 | 6144 | 1500 |
| 1 | 4 | 4 | 0.0273 | 6144 | 1500 |
| 2 | 1 | 0 | 0.1582 | 6144 | 1500 |
| 2 | 1 | 1 | 0.1543 | 6144 | 1500 |
| 2 | 1 | 2 | 0.2129 | 6144 | 1500 |
| 2 | 1 | 3 | 0.1875 | 6144 | 1500 |
| 2 | 1 | 4 | 0.1914 | 6144 | 1500 |
| 2 | 2 | 0 | 1.0000 | 6144 | 1500 |
| 2 | 2 | 1 | 1.0000 | 6144 | 1500 |
| 2 | 2 | 2 | 1.0000 | 6144 | 1500 |
| 2 | 2 | 3 | 1.0000 | 6144 | 1500 |
| 2 | 2 | 4 | 1.0000 | 6144 | 1500 |
| 2 | 3 | 0 | 0.0352 | 6144 | 1500 |
| 2 | 3 | 1 | 0.0254 | 6144 | 1500 |
| 2 | 3 | 2 | 0.0273 | 6144 | 1500 |
| 2 | 3 | 3 | 0.0312 | 6144 | 1500 |
| 2 | 3 | 4 | 0.0312 | 6144 | 1500 |
| 2 | 4 | 0 | 0.0352 | 6144 | 1500 |
| 2 | 4 | 1 | 0.0410 | 6144 | 1500 |
| 2 | 4 | 2 | 0.0371 | 6144 | 1500 |
| 2 | 4 | 3 | 0.0273 | 6144 | 1500 |
| 2 | 4 | 4 | 0.0332 | 6144 | 1500 |
| 3 | 1 | 0 | 0.1562 | 6144 | 1500 |
| 3 | 1 | 1 | 0.1660 | 6144 | 1500 |
| 3 | 1 | 2 | 0.2051 | 6144 | 1500 |
| 3 | 1 | 3 | 0.1230 | 6144 | 1500 |
| 3 | 1 | 4 | 0.1348 | 6144 | 1500 |
| 3 | 2 | 0 | 0.2363 | 6144 | 1500 |
| 3 | 2 | 1 | 0.2480 | 6144 | 1500 |
| 3 | 2 | 2 | 0.2500 | 6144 | 1500 |
| 3 | 2 | 3 | 0.2773 | 6144 | 1500 |
| 3 | 2 | 4 | 0.2246 | 6144 | 1500 |
| 3 | 3 | 0 | 1.0000 | 6144 | 1500 |
| 3 | 3 | 1 | 1.0000 | 6144 | 1500 |
| 3 | 3 | 2 | 1.0000 | 6144 | 1500 |
| 3 | 3 | 3 | 1.0000 | 6144 | 1500 |
| 3 | 3 | 4 | 1.0000 | 6144 | 1500 |
| 3 | 4 | 0 | 0.0352 | 6144 | 1500 |
| 3 | 4 | 1 | 0.0449 | 6144 | 1500 |
| 3 | 4 | 2 | 0.0332 | 6144 | 1500 |
| 3 | 4 | 3 | 0.0352 | 6144 | 1500 |
| 3 | 4 | 4 | 0.0332 | 6144 | 1500 |
| 4 | 1 | 0 | 0.1348 | 6144 | 1500 |
| 4 | 1 | 1 | 0.1875 | 6144 | 1500 |
| 4 | 1 | 2 | 0.1719 | 6144 | 1500 |
| 4 | 1 | 3 | 0.1504 | 6144 | 1500 |
| 4 | 1 | 4 | 0.1309 | 6144 | 1500 |
| 4 | 2 | 0 | 0.2402 | 6144 | 1500 |
| 4 | 2 | 1 | 0.2051 | 6144 | 1500 |
| 4 | 2 | 2 | 0.2637 | 6144 | 1500 |
| 4 | 2 | 3 | 0.2773 | 6144 | 1500 |
| 4 | 2 | 4 | 0.2207 | 6144 | 1500 |
| 4 | 3 | 0 | 0.3555 | 6144 | 1500 |
| 4 | 3 | 1 | 0.3516 | 6144 | 1500 |
| 4 | 3 | 2 | 0.3496 | 6144 | 1500 |
| 4 | 3 | 3 | 0.3633 | 6144 | 1500 |
| 4 | 3 | 4 | 0.3535 | 6144 | 1500 |
| 4 | 4 | 0 | 1.0000 | 6144 | 1500 |
| 4 | 4 | 1 | 1.0000 | 6144 | 1500 |
| 4 | 4 | 2 | 1.0000 | 6144 | 1500 |
| 4 | 4 | 3 | 1.0000 | 6144 | 1500 |
| 4 | 4 | 4 | 1.0000 | 6144 | 1500 |

**Table 23.** All 80 layer-by-hop runs. Parameter count is 6144 in every row because the inter-layer map is shared and is not a parameter; depth adds computation, not weights. The diagonal is the only set of cells above 0.99.

## Appendix G  All Fusion Evolution Curves

Section 4.13 reports final accuracies. Here the full trajectory of each arm is printed: accuracy is sampled every 10 rounds and averaged over the five seeds at each sampled round. Protocol: 32 bindings, 320 i.i.d. arrival rounds, rho = 0.999 except the fast-decay arm, epsilon = 1e-3, capacity 64, one-sided paired-CI gate. The core state is 32768 bytes at every sampled point of every run.

| arm | round | mean acc | sd | mean slots |
|---|---|---|---|---|
| fused gated | 1 | 0.0312 | 0.0000 | 1.0 |
| fused gated | 10 | 0.2687 | 0.0250 | 8.6 |
| fused gated | 20 | 0.4688 | 0.0198 | 15.0 |
| fused gated | 30 | 0.6562 | 0.0685 | 21.0 |
| fused gated | 40 | 0.7438 | 0.0459 | 23.8 |
| fused gated | 50 | 0.7812 | 0.0442 | 25.0 |
| fused gated | 60 | 0.8562 | 0.0153 | 27.4 |
| fused gated | 70 | 0.9000 | 0.0415 | 28.8 |
| fused gated | 80 | 0.9125 | 0.0415 | 29.2 |
| fused gated | 90 | 0.9500 | 0.0375 | 30.4 |
| fused gated | 100 | 0.9625 | 0.0364 | 30.8 |
| fused gated | 110 | 0.9750 | 0.0234 | 31.2 |
| fused gated | 120 | 0.9812 | 0.0250 | 31.4 |
| fused gated | 130 | 0.9812 | 0.0250 | 31.4 |
| fused gated | 140 | 0.9875 | 0.0153 | 31.6 |
| fused gated | 150 | 0.9875 | 0.0153 | 31.6 |
| fused gated | 160 | 0.9875 | 0.0153 | 31.6 |
| fused gated | 170 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 180 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 190 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 200 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 210 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 220 | 1.0000 | 0.0000 | 32.0 |

| | | | | |
|---|---|---|---|---|
| fused gated | 230 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 240 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 250 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 260 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 270 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 280 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 290 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 300 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 310 | 1.0000 | 0.0000 | 32.0 |
| fused gated | 320 | 1.0000 | 0.0000 | 32.0 |
| no consolidation | 1 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 10 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 20 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 30 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 40 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 50 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 60 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 70 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 80 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 90 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 100 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 110 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 120 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 130 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 140 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 150 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 160 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 170 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 180 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 190 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 200 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 210 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 220 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 230 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 240 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 250 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 260 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 270 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 280 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 290 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 300 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 310 | 0.0000 | 0.0000 | 0.0 |
| no consolidation | 320 | 0.0000 | 0.0000 | 0.0 |
| ungated clean | 1 | 0.0312 | 0.0000 | 1.0 |
| ungated clean | 10 | 0.2687 | 0.0250 | 8.6 |
| ungated clean | 20 | 0.4688 | 0.0198 | 15.0 |
| ungated clean | 30 | 0.6562 | 0.0685 | 21.0 |
| ungated clean | 40 | 0.7438 | 0.0459 | 23.8 |
| ungated clean | 50 | 0.7812 | 0.0442 | 25.0 |
| ungated clean | 60 | 0.8562 | 0.0153 | 27.4 |
| ungated clean | 70 | 0.9000 | 0.0415 | 28.8 |
| ungated clean | 80 | 0.9125 | 0.0415 | 29.2 |
| ungated clean | 90 | 0.9500 | 0.0375 | 30.4 |
| ungated clean | 100 | 0.9625 | 0.0364 | 30.8 |
| ungated clean | 110 | 0.9750 | 0.0234 | 31.2 |
| ungated clean | 120 | 0.9812 | 0.0250 | 31.4 |
| ungated clean | 130 | 0.9812 | 0.0250 | 31.4 |
| ungated clean | 140 | 0.9875 | 0.0153 | 31.6 |
| ungated clean | 150 | 0.9875 | 0.0153 | 31.6 |
| ungated clean | 160 | 0.9875 | 0.0153 | 31.6 |
| ungated clean | 170 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 180 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 190 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 200 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 210 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 220 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 230 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 240 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 250 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 260 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 270 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 280 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 290 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 300 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 310 | 1.0000 | 0.0000 | 32.0 |
| ungated clean | 320 | 1.0000 | 0.0000 | 32.0 |
| ungated corrupt | 1 | 0.0125 | 0.0153 | 1.0 |
| ungated corrupt | 10 | 0.1750 | 0.0153 | 8.6 |
| ungated corrupt | 20 | 0.3125 | 0.0713 | 15.2 |
| ungated corrupt | 30 | 0.4250 | 0.0580 | 20.4 |
| ungated corrupt | 40 | 0.4875 | 0.0424 | 23.4 |
| ungated corrupt | 50 | 0.5813 | 0.0580 | 25.6 |
| ungated corrupt | 60 | 0.6188 | 0.0893 | 26.8 |
| ungated corrupt | 70 | 0.6188 | 0.0459 | 28.4 |
| ungated corrupt | 80 | 0.6375 | 0.0424 | 29.8 |
| ungated corrupt | 90 | 0.6750 | 0.0319 | 30.2 |
| ungated corrupt | 100 | 0.6813 | 0.0306 | 31.0 |
| ungated corrupt | 110 | 0.6750 | 0.0424 | 31.2 |
| ungated corrupt | 120 | 0.6937 | 0.0234 | 31.4 |
| ungated corrupt | 130 | 0.6750 | 0.0468 | 31.4 |
| ungated corrupt | 140 | 0.6937 | 0.0606 | 31.6 |
| ungated corrupt | 150 | 0.7125 | 0.0500 | 32.0 |
| ungated corrupt | 160 | 0.7500 | 0.0559 | 32.0 |
| ungated corrupt | 170 | 0.7125 | 0.0800 | 32.0 |
| ungated corrupt | 180 | 0.7125 | 0.0824 | 32.0 |
| ungated corrupt | 190 | 0.7188 | 0.1296 | 32.0 |
| ungated corrupt | 200 | 0.7188 | 0.1479 | 32.0 |
| ungated corrupt | 210 | 0.7562 | 0.1389 | 32.0 |
| ungated corrupt | 220 | 0.7438 | 0.0956 | 32.0 |
| ungated corrupt | 230 | 0.7312 | 0.1019 | 32.0 |
| ungated corrupt | 240 | 0.7500 | 0.0625 | 32.0 |
| ungated corrupt | 250 | 0.6687 | 0.0673 | 32.0 |
| ungated corrupt | 260 | 0.6687 | 0.0755 | 32.0 |
| ungated corrupt | 270 | 0.6937 | 0.0415 | 32.0 |
| ungated corrupt | 280 | 0.7188 | 0.0713 | 32.0 |
| ungated corrupt | 290 | 0.7063 | 0.0153 | 32.0 |
| ungated corrupt | 300 | 0.6562 | 0.0198 | 32.0 |
| ungated corrupt | 310 | 0.6312 | 0.0234 | 32.0 |
| ungated corrupt | 320 | 0.6438 | 0.0781 | 32.0 |
| random write | 1 | 0.0000 | 0.0000 | 0.2 |
| random write | 10 | 0.1000 | 0.0125 | 4.8 |
| random write | 20 | 0.1250 | 0.0198 | 6.0 |
| random write | 30 | 0.1625 | 0.0459 | 7.8 |
| random write | 40 | 0.2500 | 0.0198 | 11.2 |
| random write | 50 | 0.3375 | 0.0125 | 15.0 |
| random write | 60 | 0.4062 | 0.0685 | 17.4 |
| random write | 70 | 0.4562 | 0.0805 | 20.6 |
| random write | 80 | 0.5062 | 0.0538 | 22.2 |
| random write | 90 | 0.5188 | 0.0468 | 22.6 |
| random write | 100 | 0.5250 | 0.0667 | 24.2 |
| random write | 110 | 0.5437 | 0.0545 | 25.0 |
| random write | 120 | 0.5750 | 0.0319 | 25.4 |
| random write | 130 | 0.5625 | 0.0625 | 26.0 |
| random write | 140 | 0.5750 | 0.0643 | 26.8 |
| random write | 150 | 0.6000 | 0.0637 | 27.0 |
| random write | 160 | 0.6375 | 0.1038 | 28.0 |
| random write | 170 | 0.6625 | 0.0871 | 28.8 |
| random write | 180 | 0.6750 | 0.0755 | 29.2 |
| random write | 190 | 0.7125 | 0.0935 | 29.4 |
| random write | 200 | 0.7000 | 0.1179 | 29.8 |
| random write | 210 | 0.7375 | 0.1000 | 30.0 |
| random write | 220 | 0.7250 | 0.0956 | 30.4 |
| random write | 230 | 0.7375 | 0.0897 | 30.8 |
| random write | 240 | 0.7500 | 0.0765 | 31.0 |
| random write | 250 | 0.6875 | 0.1046 | 31.2 |
| random write | 260 | 0.6875 | 0.1064 | 31.2 |
| random write | 270 | 0.7188 | 0.0948 | 31.4 |
| random write | 280 | 0.7063 | 0.0919 | 31.4 |
| random write | 290 | 0.7000 | 0.0805 | 31.4 |
| random write | 300 | 0.6813 | 0.0696 | 31.4 |
| random write | 310 | 0.6687 | 0.0508 | 31.4 |
| random write | 320 | 0.6875 | 0.0713 | 31.4 |
| break alignment | 1 | 0.0437 | 0.0250 | 1.0 |
| break alignment | 10 | 0.0312 | 0.0342 | 8.6 |
| break alignment | 20 | 0.0250 | 0.0125 | 15.0 |
| break alignment | 30 | 0.0312 | 0.0342 | 21.0 |
| break alignment | 40 | 0.0187 | 0.0250 | 23.8 |
| break alignment | 50 | 0.0187 | 0.0250 | 25.0 |
| break alignment | 60 | 0.0312 | 0.0198 | 27.4 |
| break alignment | 70 | 0.0312 | 0.0342 | 28.8 |
| break alignment | 80 | 0.0312 | 0.0342 | 29.2 |
| break alignment | 90 | 0.0312 | 0.0342 | 30.4 |
| break alignment | 100 | 0.0312 | 0.0395 | 30.8 |
| break alignment | 110 | 0.0125 | 0.0250 | 31.2 |
| break alignment | 120 | 0.0125 | 0.0250 | 31.4 |
| break alignment | 130 | 0.0063 | 0.0125 | 31.4 |
| break alignment | 140 | 0.0187 | 0.0153 | 31.6 |
| break alignment | 150 | 0.0125 | 0.0153 | 31.6 |
| break alignment | 160 | 0.0125 | 0.0153 | 31.6 |
| break alignment | 170 | 0.0250 | 0.0234 | 32.0 |
| break alignment | 180 | 0.0250 | 0.0234 | 32.0 |
| break alignment | 190 | 0.0187 | 0.0153 | 32.0 |
| break alignment | 200 | 0.0125 | 0.0153 | 32.0 |
| break alignment | 210 | 0.0125 | 0.0153 | 32.0 |
| break alignment | 220 | 0.0187 | 0.0250 | 32.0 |
| break alignment | 230 | 0.0312 | 0.0280 | 32.0 |
| break alignment | 240 | 0.0312 | 0.0280 | 32.0 |
| break alignment | 250 | 0.0250 | 0.0234 | 32.0 |
| break alignment | 260 | 0.0187 | 0.0250 | 32.0 |
| break alignment | 270 | 0.0187 | 0.0153 | 32.0 |
| break alignment | 280 | 0.0187 | 0.0153 | 32.0 |
| break alignment | 290 | 0.0125 | 0.0153 | 32.0 |
| break alignment | 300 | 0.0250 | 0.0364 | 32.0 |
| break alignment | 310 | 0.0250 | 0.0364 | 32.0 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| break alignment | 320 | 0.0250 | 0.0364 | 32.0 | 2 | 0 | 2 | 4 | 0.2500 | 0.0e+00 |
| rho fast | 1 | 0.0312 | 0.0000 | 1.0 | 2 | 0 | 3 | 4 | 0.2500 | 0.0e+00 |
| rho fast | 10 | 0.2687 | 0.0250 | 8.6 | 2 | 0 | 4 | 4 | 0.2500 | 0.0e+00 |
| rho fast | 20 | 0.4688 | 0.0198 | 15.0 | 2 | 0 | 5 | 4 | 0.2500 | 0.0e+00 |
| rho fast | 30 | 0.6562 | 0.0685 | 21.0 | 2 | 0 | 6 | 4 | 0.2500 | 0.0e+00 |
| rho fast | 40 | 0.7438 | 0.0459 | 23.8 | 2 | 0 | 7 | 4 | 0.2500 | 0.0e+00 |
| rho fast | 50 | 0.7812 | 0.0442 | 25.0 | 2 | 0 | 8 | 4 | 0.2500 | 0.0e+00 |
| rho fast | 60 | 0.8562 | 0.0153 | 27.4 | 2 | 1 | 1 | 4 | 0.5000 | 0.0e+00 |
| rho fast | 70 | 0.7625 | 0.0755 | 28.8 | 2 | 1 | 2 | 4 | 0.5000 | 0.0e+00 |
| rho fast | 80 | 0.5813 | 0.0580 | 29.2 | 2 | 1 | 3 | 4 | 0.5000 | 0.0e+00 |
| rho fast | 90 | 0.4188 | 0.0897 | 30.4 | 2 | 1 | 4 | 4 | 0.5000 | 0.0e+00 |
| rho fast | 100 | 0.3250 | 0.1000 | 30.8 | 2 | 1 | 5 | 4 | 0.5000 | 0.0e+00 |
| rho fast | 110 | 0.3063 | 0.0893 | 31.2 | 2 | 1 | 6 | 4 | 0.5000 | 0.0e+00 |
| rho fast | 120 | 0.2750 | 0.0824 | 31.4 | 2 | 1 | 7 | 4 | 0.5000 | 0.0e+00 |
| rho fast | 130 | 0.2375 | 0.0755 | 31.4 | 2 | 1 | 8 | 4 | 0.5000 | 0.0e+00 |
| rho fast | 140 | 0.2313 | 0.0643 | 31.6 | 2 | 2 | 1 | 4 | 1.0000 | 0.0e+00 |
| rho fast | 150 | 0.2000 | 0.0755 | 31.6 | 2 | 2 | 2 | 4 | 1.0000 | 0.0e+00 |
| rho fast | 160 | 0.1750 | 0.0545 | 31.6 | 2 | 2 | 3 | 4 | 1.0000 | 0.0e+00 |
| rho fast | 170 | 0.1562 | 0.0656 | 32.0 | 2 | 2 | 4 | 4 | 1.0000 | 0.0e+00 |
| rho fast | 180 | 0.1562 | 0.0442 | 32.0 | 2 | 2 | 5 | 4 | 1.0000 | 0.0e+00 |
| rho fast | 190 | 0.1875 | 0.0523 | 32.0 | 2 | 2 | 6 | 4 | 1.0000 | 0.0e+00 |
| rho fast | 200 | 0.1500 | 0.0750 | 32.0 | 2 | 2 | 7 | 4 | 1.0000 | 0.0e+00 |
| rho fast | 210 | 0.1500 | 0.0871 | 32.0 | 2 | 2 | 8 | 4 | 1.0000 | 0.0e+00 |
| rho fast | 220 | 0.1500 | 0.0696 | 32.0 | 3 | 0 | 1 | 8 | 0.1250 | 0.0e+00 |
| rho fast | 230 | 0.1250 | 0.0656 | 32.0 | 3 | 0 | 2 | 8 | 0.1250 | 0.0e+00 |
| rho fast | 240 | 0.1375 | 0.0643 | 32.0 | 3 | 0 | 3 | 8 | 0.1250 | 0.0e+00 |
| rho fast | 250 | 0.1187 | 0.0306 | 32.0 | 3 | 0 | 4 | 8 | 0.1250 | 0.0e+00 |
| rho fast | 260 | 0.1125 | 0.0250 | 32.0 | 3 | 0 | 5 | 8 | 0.1250 | 0.0e+00 |
| rho fast | 270 | 0.1375 | 0.0153 | 32.0 | 3 | 0 | 6 | 8 | 0.1250 | 0.0e+00 |
| rho fast | 280 | 0.1250 | 0.0280 | 32.0 | 3 | 0 | 7 | 8 | 0.1250 | 0.0e+00 |
| rho fast | 290 | 0.1313 | 0.0364 | 32.0 | 3 | 0 | 8 | 8 | 0.1250 | 0.0e+00 |
| rho fast | 300 | 0.1437 | 0.0250 | 32.0 | 3 | 1 | 1 | 8 | 0.2500 | 0.0e+00 |
| rho fast | 310 | 0.1250 | 0.0342 | 32.0 | 3 | 1 | 2 | 8 | 0.2500 | 0.0e+00 |
| rho fast | 320 | 0.1313 | 0.0234 | 32.0 | 3 | 1 | 3 | 8 | 0.2500 | 0.0e+00 |
| | | | | | 3 | 1 | 4 | 8 | 0.2500 | 0.0e+00 |
| | | | | | 3 | 1 | 5 | 8 | 0.2500 | 0.0e+00 |
| | | | | | 3 | 1 | 6 | 8 | 0.2500 | 0.0e+00 |
| | | | | | 3 | 1 | 7 | 8 | 0.2500 | 0.0e+00 |
| | | | | | 3 | 1 | 8 | 8 | 0.2500 | 0.0e+00 |
| | | | | | 3 | 3 | 1 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 2 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 3 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 4 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 5 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 6 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 7 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 8 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 4 | 0 | 1 | 16 | 0.0625 | 0.0e+00 |
| | | | | | 4 | 0 | 2 | 16 | 0.0625 | 0.0e+00 |
| | | | | | 4 | 0 | 3 | 16 | 0.0625 | 0.0e+00 |
| | | | | | 4 | 0 | 4 | 16 | 0.0625 | 0.0e+00 |
| | | | | | 4 | 0 | 5 | 16 | 0.0625 | 0.0e+00 |
| | | | | | 4 | 0 | 6 | 16 | 0.0625 | 0.0e+00 |
| | | | | | 4 | 0 | 7 | 16 | 0.0625 | 0.0e+00 |
| | | | | | 4 | 0 | 8 | 16 | 0.0625 | 0.0e+00 |
| | | | | | 4 | 1 | 1 | 16 | 0.1250 | 0.0e+00 |
| | | | | | 4 | 1 | 2 | 16 | 0.1250 | 0.0e+00 |
| | | | | | 4 | 1 | 3 | 16 | 0.1250 | 0.0e+00 |
| | | | | | 4 | 1 | 4 | 16 | 0.1250 | 0.0e+00 |
| | | | | | 4 | 1 | 5 | 16 | 0.1250 | 0.0e+00 |
| | | | | | 4 | 1 | 6 | 16 | 0.1250 | 0.0e+00 |
| | | | | | 4 | 1 | 7 | 16 | 0.1250 | 0.0e+00 |
| | | | | | 4 | 1 | 8 | 16 | 0.1250 | 0.0e+00 |
| | | | | | 4 | 4 | 1 | 16 | 1.0000 | 0.0e+00 |
| | | | | | 4 | 4 | 2 | 16 | 1.0000 | 0.0e+00 |
| | | | | | 4 | 4 | 3 | 16 | 1.0000 | 0.0e+00 |
| | | | | | 4 | 4 | 4 | 16 | 1.0000 | 0.0e+00 |
| | | | | | 4 | 4 | 5 | 16 | 1.0000 | 0.0e+00 |
| | | | | | 4 | 4 | 6 | 16 | 1.0000 | 0.0e+00 |
| | | | | | 4 | 4 | 7 | 16 | 1.0000 | 0.0e+00 |
| | | | | | 4 | 4 | 8 | 16 | 1.0000 | 0.0e+00 |
| | | | | | 1 | 1 | 1 | 2 | 1.0000 | 0.0e+00 |
| | | | | | 1 | 1 | 2 | 2 | 0.0000 | 0.0e+00 |
| | | | | | 1 | 1 | 3 | 2 | 0.0000 | 0.0e+00 |
| | | | | | 1 | 1 | 4 | 2 | 0.0000 | 0.0e+00 |
| | | | | | 1 | 1 | 5 | 2 | 0.0586 | 0.0e+00 |
| | | | | | 2 | 2 | 1 | 4 | 0.0000 | 0.0e+00 |
| | | | | | 2 | 2 | 2 | 4 | 1.0000 | 0.0e+00 |
| | | | | | 2 | 2 | 3 | 4 | 0.0000 | 0.0e+00 |
| | | | | | 2 | 2 | 4 | 4 | 0.0000 | 0.0e+00 |
| | | | | | 2 | 2 | 5 | 4 | 0.0625 | 0.0e+00 |
| | | | | | 3 | 3 | 1 | 8 | 0.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 2 | 8 | 0.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 3 | 8 | 1.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 4 | 8 | 0.0000 | 0.0e+00 |
| | | | | | 3 | 3 | 5 | 8 | 0.0801 | 0.0e+00 |
| | | | | | 4 | 4 | 1 | 16 | 0.0000 | 0.0e+00 |

**Table 24.** All 231 arm-round cells. The fused arm is monotone on all five seeds; the ungated corrupted arm rises then fluctuates as later writes overwrite earlier correct registrations; the fast-decay arm loses slots as SE5's finite-precision capacity is exhausted.

| comparison | paired diff | CI95 low | CI95 high | all pos |
|---|---|---|---|---|
| no consolidation | +1.0000 | +1.0000 | +1.0000 | yes |
| ungated clean | +0.0000 | +0.0000 | +0.0000 | no |
| ungated corrupt | +0.3563 | +0.2875 | +0.4250 | yes |
| random write | +0.3125 | +0.2500 | +0.3750 | yes |
| break alignment | +0.9750 | +0.9375 | +1.0000 | yes |
| rho fast | +0.8688 | +0.8500 | +0.8875 | yes |

**Table 25.** All 6 paired bootstrap comparisons, 10000 resamples, five seeds. The ungated clean arm ties exactly: with no corruption the gate is irrelevant, which is the measured form of SE3's scope, not a failure of the comparison.

# Appendix H  Exhaustive Information-Bound Enumeration

Theorem A is checked against the complete enumeration of the binary-chain fibre space, not against a sample. Every (m, B) pair below is enumerated over all 2^m role assignments with bit-identical visible inputs. The depth scan then verifies that stacking 1-8 static-LTI layers produces bit-identical outputs across compatible role assignments. These are finite exhaustive objects; they certify non-vacuity, they are not the proof.

| m | B | answer classes | fibres | acc | predicted |
|---|---|---|---|---|---|
| 1 | 0 | 2 | 1 | 0.5000 | 0.5000 |
| 1 | 1 | 2 | 2 | 1.0000 | 1.0000 |
| 1 | 2 | 2 | 2 | 1.0000 | 1.0000 |
| 2 | 0 | 4 | 1 | 0.2500 | 0.2500 |
| 2 | 1 | 4 | 2 | 0.5000 | 0.5000 |
| 2 | 2 | 4 | 4 | 1.0000 | 1.0000 |
| 2 | 3 | 4 | 4 | 1.0000 | 1.0000 |
| 3 | 0 | 8 | 1 | 0.1250 | 0.1250 |
| 3 | 1 | 8 | 2 | 0.2500 | 0.2500 |
| 3 | 2 | 8 | 4 | 0.5000 | 0.5000 |
| 3 | 3 | 8 | 8 | 1.0000 | 1.0000 |
| 3 | 4 | 8 | 8 | 1.0000 | 1.0000 |
| 4 | 0 | 16 | 1 | 0.0625 | 0.0625 |
| 4 | 1 | 16 | 2 | 0.1250 | 0.1250 |
| 4 | 2 | 16 | 4 | 0.2500 | 0.2500 |
| 4 | 3 | 16 | 8 | 0.5000 | 0.5000 |
| 4 | 4 | 16 | 16 | 1.0000 | 1.0000 |
| 4 | 5 | 16 | 16 | 1.0000 | 1.0000 |

**Table 26.** All 18 (m, B) enumeration cells. Accuracy equals the predicted upper bound 2^B / |A| in every row and reaches 1 exactly at B = m.

| m | B | layers | \|A\| | acc | output dev |
|---|---|---|---|---|---|
| 2 | 0 | 1 | 4 | 0.2500 | 0.0e+00 |

| | | | | | |
|---|---|---|---|---|---|
| 4 | 4 | 2 | 16 | 0.0000 | 0.0e+00 |
| 4 | 4 | 3 | 16 | 0.0000 | 0.0e+00 |
| 4 | 4 | 4 | 16 | 1.0000 | 0.0e+00 |
| 4 | 4 | 5 | 16 | 0.0898 | 0.0e+00 |
| 2 | 2 | 3 | 4 | 1.0000 | 0.0e+00 |
| 2 | 2 | 4 | 4 | 1.0000 | 0.0e+00 |
| 3 | 3 | 4 | 8 | 1.0000 | 0.0e+00 |
| 3 | 3 | 5 | 8 | 1.0000 | 0.0e+00 |
| 4 | 4 | 5 | 16 | 1.0000 | 0.0e+00 |
| 4 | 4 | 6 | 16 | 1.0000 | 0.0e+00 |

**Table 27.** All 98 depth-scan cells. Accuracy and output deviation are invariant to layer count, which is the empirical form of Theorem A(e): depth cannot manufacture the missing bits.

| carrier | bits / token | acc | Toeplitz | run-time tag |
|---|---|---|---|---|
| explicit tag | 1 | 1.0000 | yes | yes |
| codebook partition | 1 | 1.0000 | yes | no |

**Table 28.** All 3 bit carriers. The position-format-prior accuracy 0.2969 is a SINGLE-SEED measurement on the alternating layout under that carrier's support assumption; it is a DIFFERENT configuration from the frozen five-seed parity-gapped value in Table 10 and is never quoted as a conclusion elsewhere. The position format prior is a support restriction, not a per-token channel, and is analysed through A(v) rather than through B; its effective bit count under arbitrary filler is zero.

| hops | layers | acc | per-layer accuracies |
|---|---|---|---|
| 1 | 1 | 1.0000 | 1.00 |
| 1 | 2 | 0.0000 | 1.00 0.00 |
| 1 | 3 | 0.0000 | 1.00 0.00 0.00 |
| 1 | 4 | 0.0000 | 1.00 0.00 0.00 0.00 |
| 1 | 5 | 0.0645 | 1.00 0.00 0.00 0.00 0.06 |
| 2 | 1 | 0.0000 | 0.00 |
| 2 | 2 | 1.0000 | 0.00 1.00 |
| 2 | 3 | 0.0000 | 0.00 1.00 0.00 |
| 2 | 4 | 0.0000 | 0.00 1.00 0.00 0.00 |
| 2 | 5 | 0.0684 | 0.00 1.00 0.00 0.00 0.07 |
| 3 | 1 | 0.0000 | 0.00 |
| 3 | 2 | 0.0000 | 0.00 0.00 |
| 3 | 3 | 1.0000 | 0.00 0.00 1.00 |
| 3 | 4 | 0.0000 | 0.00 0.00 1.00 0.00 |
| 3 | 5 | 0.0879 | 0.00 0.00 1.00 0.00 0.09 |
| 4 | 1 | 0.0000 | 0.00 |
| 4 | 2 | 0.0000 | 0.00 0.00 |
| 4 | 3 | 0.0000 | 0.00 0.00 0.00 |
| 4 | 4 | 1.0000 | 0.00 0.00 0.00 1.00 |
| 4 | 5 | 0.0918 | 0.00 0.00 0.00 1.00 0.09 |

**Table 29.** All 20 zero-training layer-by-hop cells with the full per-layer accuracy vector. The last-layer read-out is why H_L > H overshoots fail: the correct intermediate value is computed at layer H but is not the layer that is read out.

# Appendix I Complete Numeric Provenance and Witness Detail

Every numeric value quoted anywhere in the paper is listed below with its statistic, seed count, sample count and the object it was measured on. This table exists so that no number has to be taken on trust and no earlier drifted value can re-enter unnoticed. All entries are finite-sample quantities unless the statistic column says otherwise; none is a theorem.

| quantity | value | statistic | seeds / n | object |
|---|---|---|---|---|
| chance (this task) | 0.0312 | definition | - | 1/V_K |
| H=2, 2 layers, 1500 steps | 1.0000 | mean / sd 0.0000 | 5 / 512 | architecture stack |
| H=2, 1 layer, 1500 steps | 0.1809 | mean / sd 0.0219 | 5 / 512 | architecture stack |
| H=2 paired difference | +0.8191 | paired bootstrap | 5 | two-layer minus one-layer |
| paired 95%% CI | [+0.8008, +0.8375] | 10000 resamples | 5 | same pair |
| first step >= 0.99 | 150 | max over seeds | 5 | codebook-fixed interface |
| legacy random interface, 1500 | 0.2492 | mean | 5 | learnable inter-layer matrix |
| SE2 predicted E[T_cover] | 129.87 | analytic n H_n / (1-beta) | - | n=32, beta~0 |
| SE2 measured mean cover | 130.00 | mean over seeds | 5 | fused gated arm |
| SE2 relative gap | 0.0987% | derived (mean coincidence, NOT precision) | - | prediction vs measurement |
| SE2 cover 95% t-CI | [82.72, 177.28] | t(4), sample SD 38.08 | 5 | per-seed first full cover |
| SE3 exact E[acc] | 0.709375 | analytic | - | p=0.30, V_K=32 |
| SE3 enumeration mean | 0.709102 | mean over 640000 key trials | 20000 runs | exhaustive per-key |
| SE3 Hoeffding 4-sigma | [0.7071, 0.7116] | concentration | - | same |
| SE4 bits consumed | 500 | analytic | - | T=100, w=1, b=5 |
| SE4 CT7 bound | 500 | analytic upper bound | - | same |
| SE5 K* at rho=0.999 | 6904 | floor(log eps / log rho) | - | eps=1e-3 |
| core state bytes | 32768 | exact | - | d x d fp64, all runs |
| fused final accuracy | 1.0000 | mean / sd 0.0000 | 5 / 512 | 320 rounds |
| ungated corrupted | 0.6438 | mean / sd 0.0781 | 5 / 512 | p=0.30, no gate |
| break alignment | 0.0250 | mean / sd 0.0364 | 5 / 512 | non-codebook registration |
| fast decay | 0.1313 | mean / sd 0.0234 | 5 / 512 | rho=0.90 |
| SE1 max on-target error | 4.4e-16 | max over 200 instances | 200 | orthogonality identity |
| SE1 max off-target error | 8.5e-17 | max over 200 instances | 200 | same |
| FFT/IIR duality (final state) | 2.89e-16 | max over probes | - | real-field equivalence |
| Toeplitz deviation (final) | 0 | exact | - | static recurrence |

**Table 30.** All 26 provenance rows. Values from different probes are never averaged together; the FFT/IIR row is a maximum over separately listed probes, not a mixture.

| witness | measured value | what it certifies |
|---|---|---|
| evolution all invariants | pass | consistency on final weights |
| evolution monotone | pass | consistency on final weights |
| fused above 0.90 | pass | consistency on final weights |
| all key paired positive | pass | consistency on final weights |
| core state bytes constant across rounds | pass | consistency on final weights |
| slot count bounded by capacity | pass | consistency on final weights |
| SE5 capacity predicts rho fast loss | pass | consistency on final weights |
| ablation clean gate configs saturate | pass | consistency on final weights |
| ablation noisy gate necessary | pass | consistency on final weights |
| ablation rho decay matches SE5 | pass | consistency on final weights |
| SE2 prediction matches measurement | pass | consistency on final weights |
| SE3 concentration holds | pass | consistency on final weights |
| SE3 enumeration within 4sigma | pass | consistency on final weights |
| no gradient through retrieval | pass | consistency on final weights |
| no NxN score | pass | consistency on final weights |
| no kv cache | pass | consistency on final weights |
| SE2 prediction gap | 0.0987% | coupon-collector bound is non-vacuous |
| SE3 concentration | inside 4-sigma | exact expectation, not an upper bound |
| SE4 equality | 500 = 500 bits | no contradiction with CT7 |
| SE5 capacity | 6904 vs 65 | decay predicts the loss |

**Table 31.** Witness detail for the non-analytic checks. A false here would be a finding to fix, not a crashed run.

| hops | decoding | n | mean \|A\| | max log2\|A\| | frac \|A\|>1 |
|---|---|---|---|---|---|
| 1 | flat | 1280 | 1.00 | 0.00 | 0.00 |
| 1 | codebook | 1280 | 1.00 | 0.00 | 0.00 |
| 2 | flat | 1280 | 1.52 | 1.00 | 0.52 |
| 2 | codebook | 1280 | 1.00 | 0.00 | 0.00 |
| 3 | flat | 1280 | 2.36 | 2.00 | 0.75 |
| 3 | codebook | 1280 | 1.00 | 0.00 | 0.00 |
| 4 | flat | 1280 | 3.25 | 2.32 | 0.97 |
| 4 | codebook | 1280 | 1.00 | 0.00 | 0.00 |

**Table 32.** Compatible-answer-set enumeration for the original content-chain task. The bound B >= log2 |A(v)| is sufficient but not tight on this task; tightness belongs to the binary-chain witness of Appendix H. This distinction is stated in Section 4.3 and is not hidden here.

# Appendix J Per-Seed Raw Curves at Key Steps

Appendix F averages over seeds at each step; this appendix prints the individual trajectories so that no seed-level failure can be hidden by a mean. Each row is one seed of one arm at one evaluation step. The five evaluation steps are 0, 150, 400, 800 and 1500, taken from the same 1500-step runs used in Section 4.12. Loss is the held-out cross-entropy on a fixed 64-instance probe batch; accuracy is on the full 512-instance held-out set.

| arm | seed | step | acc | loss |
|---|---|---|---|---|
| arch codebook M | 0 | 0 | 0.0273 | 3.4683 |
| arch codebook M | 0 | 150 | 1.0000 | 0.0655 |
| arch codebook M | 0 | 1500 | 1.0000 | 0.0017 |
| arch codebook M | 1 | 0 | 0.0273 | 3.4534 |
| arch codebook M | 1 | 150 | 1.0000 | 0.0608 |

| | | | | |
|---|---|---|---|---|
| arch codebook M | 1 | 1500 | 1.0000 | 0.0015 |
| arch codebook M | 2 | 0 | 0.0234 | 3.4731 |
| arch codebook M | 2 | 150 | 1.0000 | 0.0597 |
| arch codebook M | 2 | 1500 | 1.0000 | 0.0014 |
| arch codebook M | 3 | 0 | 0.0371 | 3.4595 |
| arch codebook M | 3 | 150 | 1.0000 | 0.0751 |
| arch codebook M | 3 | 1500 | 1.0000 | 0.0024 |
| arch codebook M | 4 | 0 | 0.0176 | 3.4635 |
| arch codebook M | 4 | 150 | 1.0000 | 0.0756 |
| arch codebook M | 4 | 1500 | 1.0000 | 0.0021 |
| arch analytic witness | 0 | 0 | 1.0000 | 0.0000 |
| arch analytic witness | 1 | 0 | 1.0000 | 0.0000 |
| arch analytic witness | 2 | 0 | 1.0000 | 0.0000 |
| arch analytic witness | 3 | 0 | 1.0000 | 0.0000 |
| arch analytic witness | 4 | 0 | 1.0000 | 0.0000 |
| legacy learnable interface rando | 0 | 0 | 0.0332 | 3.4666 |
| legacy learnable interface rando | 0 | 150 | 0.2852 | 3.8409 |
| legacy learnable interface rando | 0 | 1500 | 0.2617 | 4.9533 |
| legacy learnable interface rando | 1 | 0 | 0.0293 | 3.4664 |
| legacy learnable interface rando | 1 | 150 | 0.2637 | 4.0069 |
| legacy learnable interface rando | 1 | 1500 | 0.2129 | 5.6684 |
| legacy learnable interface rando | 2 | 0 | 0.0371 | 3.4607 |
| legacy learnable interface rando | 2 | 150 | 0.2207 | 3.6956 |
| legacy learnable interface rando | 2 | 1500 | 0.2090 | 4.7915 |
| legacy learnable interface rando | 3 | 0 | 0.0215 | 3.4631 |
| legacy learnable interface rando | 3 | 150 | 0.3223 | 3.6256 |
| legacy learnable interface rando | 3 | 1500 | 0.2734 | 6.1103 |
| legacy learnable interface rando | 4 | 0 | 0.0293 | 3.4697 |
| legacy learnable interface rando | 4 | 150 | 0.3145 | 3.8663 |
| legacy learnable interface rando | 4 | 1500 | 0.2891 | 4.8512 |
| legacy learnable interface fixed | 0 | 0 | 0.0391 | 3.4783 |
| legacy learnable interface fixed | 0 | 150 | 1.0000 | 0.1207 |
| legacy learnable interface fixed | 0 | 1500 | 1.0000 | 0.0019 |
| legacy learnable interface fixed | 1 | 0 | 0.0215 | 3.4747 |
| legacy learnable interface fixed | 1 | 150 | 1.0000 | 0.0738 |
| legacy learnable interface fixed | 1 | 1500 | 1.0000 | 0.0017 |
| legacy learnable interface fixed | 2 | 0 | 0.0312 | 3.4700 |
| legacy learnable interface fixed | 2 | 150 | 1.0000 | 0.0581 |
| legacy learnable interface fixed | 2 | 1500 | 1.0000 | 0.0015 |
| legacy learnable interface fixed | 3 | 0 | 0.0449 | 3.4793 |
| legacy learnable interface fixed | 3 | 150 | 1.0000 | 0.1008 |
| legacy learnable interface fixed | 3 | 1500 | 1.0000 | 0.0027 |
| legacy learnable interface fixed | 4 | 0 | 0.0566 | 3.4705 |
| legacy learnable interface fixed | 4 | 150 | 1.0000 | 0.0731 |
| legacy learnable interface fixed | 4 | 1500 | 1.0000 | 0.0021 |
| legacy trainable near-swap | 0 | 0 | 0.0234 | 3.4641 |
| legacy trainable near-swap | 0 | 150 | 1.0000 | 0.0569 |
| legacy trainable near-swap | 0 | 1500 | 1.0000 | 0.0016 |
| legacy trainable near-swap | 1 | 0 | 0.0410 | 3.4781 |
| legacy trainable near-swap | 1 | 150 | 1.0000 | 0.0623 |
| legacy trainable near-swap | 1 | 1500 | 1.0000 | 0.0016 |
| legacy trainable near-swap | 2 | 0 | 0.0254 | 3.4697 |
| legacy trainable near-swap | 2 | 150 | 1.0000 | 0.0496 |
| legacy trainable near-swap | 2 | 1500 | 1.0000 | 0.0013 |
| legacy trainable near-swap | 3 | 0 | 0.0215 | 3.4606 |
| legacy trainable near-swap | 3 | 150 | 1.0000 | 0.1082 |
| legacy trainable near-swap | 3 | 1500 | 1.0000 | 0.0025 |
| legacy trainable near-swap | 4 | 0 | 0.0078 | 3.4713 |
| legacy trainable near-swap | 4 | 150 | 1.0000 | 0.0681 |
| legacy trainable near-swap | 4 | 1500 | 1.0000 | 0.0022 |

**Table 33.** All 65 per-seed curve values. The legacy random-interface rows stay between 0.21 and 0.29 at every step on every seed, far from this task's 1/32 chance and never above 0.30; the codebook-fixed rows reach 1.0 by step 150 on every seed and never regress. The analytic witness is exact at step 0. No seed was excluded and no value was rounded from a different run.

| arm | seed | first full cover | committed | core bytes |
|---|---|---|---|---|
| fused gated | 0 | 170 | 49 | 32768 |
| fused gated | 1 | 100 | 55 | 32768 |
| fused gated | 2 | 90 | 47 | 32768 |
| fused gated | 3 | 120 | 53 | 32768 |
| fused gated | 4 | 170 | 54 | 32768 |
| no consolidation | 0 | none | 0 | 32768 |
| no consolidation | 1 | none | 0 | 32768 |
| no consolidation | 2 | none | 0 | 32768 |
| no consolidation | 3 | none | 0 | 32768 |
| no consolidation | 4 | none | 0 | 32768 |
| ungated clean | 0 | 170 | 320 | 32768 |
| ungated clean | 1 | 100 | 320 | 32768 |
| ungated clean | 2 | 90 | 320 | 32768 |
| ungated clean | 3 | 120 | 320 | 32768 |
| ungated clean | 4 | 170 | 320 | 32768 |
| ungated corrupt | 0 | none | 320 | 32768 |
| ungated corrupt | 1 | none | 320 | 32768 |
| ungated corrupt | 2 | none | 320 | 32768 |
| ungated corrupt | 3 | none | 320 | 32768 |
| ungated corrupt | 4 | none | 320 | 32768 |
| random write | 0 | none | 154 | 32768 |
| random write | 1 | none | 153 | 32768 |
| random write | 2 | none | 154 | 32768 |
| random write | 3 | none | 155 | 32768 |
| random write | 4 | none | 155 | 32768 |
| break alignment | 0 | none | 49 | 32768 |
| break alignment | 1 | none | 55 | 32768 |
| break alignment | 2 | none | 47 | 32768 |
| break alignment | 3 | none | 53 | 32768 |
| break alignment | 4 | none | 54 | 32768 |
| rho fast | 0 | none | 49 | 32768 |
| rho fast | 1 | none | 55 | 32768 |
| rho fast | 2 | none | 47 | 32768 |
| rho fast | 3 | none | 53 | 32768 |
| rho fast | 4 | none | 54 | 32768 |

**Table 34.** All 35 fusion runs with their first full-cover round, number of committed experiences and core state size. The core state is constant across every arm, every seed and every round; only the bounded slot metadata changes. These are the individual values behind Table 16 and Table 18.

**End of artifact.** No further numeric claims appear anywhere in the submission.